



ANA-specific antibodies, ANA patterns, anti-ds-DNA results, and clinical diagnosis: a laboratory and clinical audit

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Abstract

The diagnosis of systemic autoimmune diseases (SAID) is challenging, due to overlapping features with other non-immune disorders. Anti-nuclear antibodies (ANA)/anti-cellular antibodies are the sensitive screening tests but anti-double-stranded-deoxyribonucleic acid-antibody (anti-ds-DNA) and ANA-specific antibodies are specific for SAID. We aimed to look at ANA-specific antibodies in our patients and correlated them with ANA patterns, anti-ds-DNA, and clinical diagnosis for proper interpretation and better patient management cost-effectively. A retrospective data analysis of 641 patients was done (1st of February 2019 to 31st of July 2021) who were tested for ANA-specific antibodies at the Immunology Department of Indus Hospital and Health Network. ANA and anti-ds-DNA results and clinical diagnosis were also analyzed for ANA-specific antibody-positive patients. Descriptive data were presented in mean $\pm$ standard deviation and frequency percentages whereas inferential data were analyzed with a *chi-square* test for association between ANA-specific antibodies status, ANA, anti-ds-DNA, and clinical features. ANA-specific antibodies test revealed positivity for at least one autoantibody in 245 (38.2%) patients. Of these, ANA was tested in 206 patients reactive for ANA-specific antibodies and found positive in 195 (95%) as compared to negative (<0.001). Speckled and homogenous were predominant ANA patterns in ANA-specific antibody-positives (56% and 42% respectively). Multiple ANA patterns were found in 18 patients most commonly with systemic lupus erythematosus (SLE) and mixed connective tissue disorder (MCTD). Anti-SSA were the most common ANA-specific antibodies (50%) and were mostly found in sera with speckled (61/97) and homogenous (38/97) patterns and associated mostly with SLE (48%) and Sjogren's syndrome (86%). Among ANA-negative patients, anti-SSA were the most common antibodies ($n=5$). Anti-ds-DNA was found in 66% of SLE patients along with another ANA-specific antibody. This study showed that testing for ANA-specific antibodies cannot be gated on ANA patterns. Also, there is a redundancy of these antibodies with various clinical diagnoses. Moreover, they are useful in making a diagnosis in ANA-negative patients as well with clinical suspicion.

Keywords Systemic autoimmune diseases · Antigen-specific anti-nuclear antibodies · Anti-nuclear antibodies · ANA patterns · Anti-ds-DNA · Anti-cellular antibodies

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Introduction

Systemic autoimmune diseases (SAID) or connective tissue disorders (CTD) comprise 5–10% of the burden of chronic debilitating diseases causing morbidity and mortality. Over a hundred SAID are already known. The clinical features of SAID overlap with infectious diseases or other non-immune-mediated disorders. This makes the diagnosis of these disorders challenging. Furthermore, these various SAID often coexist, imposing more challenges to diagnosing and managing these disorders. A battery of immunological and non-immunological tests is required for a conclusive diagnosis of SAID [1–3].

SAID can be organ-specific or non-organ-specific. Anti-nuclear antibodies (ANA) are the hallmark of many non-organ-specific systemic autoimmune diseases such as systemic lupus erythematosus (SLE), Sjogren's syndrome (SS), systemic sclerosis (SSc), and mixed connective tissue disorders (MCTD). ANA are preferentially detected by indirect immunofluorescent assay (IFA) which also tells us about the patterns and titers. The ANA patterns are also pleomorphic and some of them can represent more than one disease-specific autoantibody [4]. Also, ANA can be present non-specifically in around 10–30% of healthy individuals at low titers (3,5). Other disease-specific autoantibody detections for further SAID classification are required including anti-ds-deoxyribonucleic acid (ds-DNA) and ANA-specific antibodies [5]. These autoantibodies comprise a panel of antibodies with mostly overlapping specificities. For example, anti-SSA can be found in SLE, SS, and SSc. Therefore, their interpretation can only be made in association with clinical features [6]. Historically, these antibodies were called anti-extractable nuclear antigens and were available with limited panels. Recently various extended panels have become available including novel ANA-specific antibodies with various clinical relevance [7].

There is enormous data available on the utility and interpretation of ANA/anti-cellular antibodies (ACA), anti-ANA-specific autoantibodies, and anti-ds-DNA. However, comprehensive data are lacking on the relation of various ANA patterns, ANA-specific antibodies, and clinical features in our local population [8, 9]. Moreover, there is generally a lack of awareness regarding the requisition and interpretation of these antibodies to diagnose SAID.

We aimed to do an audit of ANA-specific antibodies and correlate them with ANA/ACA, anti-ds-DNA results, and clinical features in our local patient population. This may help to guide clinicians to utilize these autoantibodies for the diagnosis of SAID and their management cost-effectively.

Material and methods

The study was conducted in the Department of Immunology at the Indus Hospital and Health Network (IHHN), Karachi, Pakistan. Retrospective data analysis of 641 patients was done from the 1st of February 2019 to the 31st of July 2021 who were tested for ANA-specific antibodies. ANA and anti-ds-DNA results were also analyzed for these patients if requested. For those patients who had a positive ANA-specific antibody, clinical features were also analyzed. The data were retrieved from the electronic medical record after approval of the Institutional review board at IHHN.

ANA-specific antibodies were performed by immunoblot assay using a commercially available kit (ANA Profile et Mi-2 et Ku (IgG) EUROLINE, Euroimmun AG,

Switzerland), detecting Mi-2, Ku, snRNP/Sm, Sm, SSA, Ro-52, SSB, Scl-70, PM-Scl, Jo-1, CENP B, PCNA, nucleosomes, histones, and ribosomal P proteins. We did not analyze AMA M2 as these are not considered ANA-specific antibodies. The cutoff titer was 1:10 as per the manufacturer's instructions. The bands were scanned and analyzed using EUROLinescan software and also visualized. Depending upon the band intensities, the results were interpreted as negative (0), positive ((+), +, ++, +++) according to the manufacturer's instructions, and indeterminate if there was a discrepancy in the interpretation of weak band intensities as seen on scanned reports versus visual inspection. ANA was performed by indirect immunofluorescent assay (IFA) using commercially available kits (IIFT-Mosaic: Basic Profile 3, Euroimmun, Switzerland; from February 2019 to March 2021 for ANA; and AESKUSLIDES ANA-HEp-2, AESKU, Germany; from April 2021 to July 2021). ANA results were interpreted as per the manufacturer's instructions. ANA titers of $\geq 1:100$ (by Euroimmun assay) and $\geq 1:80$ (by AESKU assay kit) were taken as positive. We analyzed the results of anti-ds-DNA performed by indirect IFA using commercially available kits (AESKUSLIDES[®]nDNA (*Crithidia Luciliae*) Test Kits, AESKU, Germany). The results were reported qualitatively with titers $\geq 1:10$ as positive. We did not analyze anti-ds-DNA bands on ANA Profile et Mi-2 et Ku by Euroimmun.

Descriptive data presented in mean $\pm$ standard deviation and frequency percentages whereas inferential data analyzed with a chi-square test for association between ANA-specific antibodies status, ANA, anti-ds-DNA, and clinical features. The statistical analysis was performed using IBM SPSS 24.0 for windows with the statistical significance threshold set at $P < 0.05$.

The study was conducted according to the standards of the Declaration of Helsinki and after approval from the institutional review board (IRB) at IHHN (IHHN_IRB_2021_08_08).

Results

The mean age of the patients was 32 ± 13 years. There were 531 (82.8%) females. Of 641 patients with a request for ANA-specific antibodies, ANA results were available for 577 (90%) and were positive in 454 patients (78.7%) with speckled (n : 288; 63.4%), homogenous (n : 150; 33%), nucleolar (n : 34; 7.5%), centromeric (n : 4; 0.9%), laminar (n : 14; 3.1%), mitotic spindle (n : 10; 2.2%), mid-body (n : 7; 1.5%), Numa-like (n : 1; 0.2%), dense fine speck (n : 6; 1.3%) patterns. Anti-ds-DNA was done for ANA-positive patients and found positive in 101 (23%) (Fig. 1). ANA-specific antibodies were positive for at least one or more autoantibodies in 245 (38.2%) and indeterminate for one or more antibodies

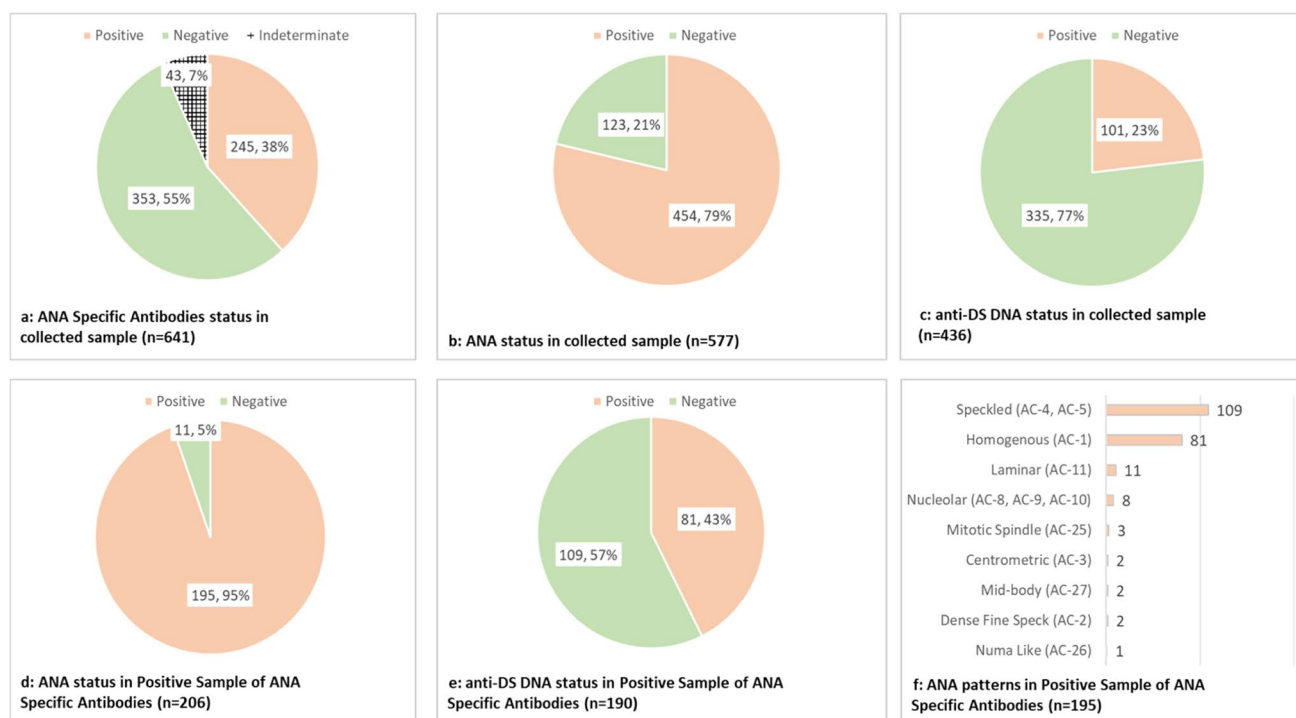


Fig. 1 Presentation of ANA-specific antibodies, ANA, anti-ds-DNA, and ANA patterns in study sample

in 43 (6.7%) patients. of these The data were analyzed for clinical diagnosis of patients positive for ANA-specific antibodies ($n = 245$).

Among the ANA-specific antibody-positive cohort, ANA was tested in 206 patients and found positive in 195 (95%), while negative in 11 ($P = 0.000$). The distribution of ANA patterns in ANA-specific antibody-positive sera is shown in Fig. 1. Overall, the most prevalent ANA pattern in ANA-specific antibody-positive samples was speckled found in 63% followed by homogenous in 54%. In 18 (9%) patients' samples, multiple ANA patterns vs. single pattern was identified ($P = 0.086$).

ANA-specific antibodies and ANA patterns

Figure 2 is showing the distribution of ANA-specific antibody results. Table 1 is showing the correlation between ANA patterns and ANA-specific antibodies. Anti-SSA antibodies were the most prevalent (50%) ANA-specific antibodies both overall and in the ANA-positive cohort and correlated mostly with speckled patterns.

Among ANA-negative (AC-0) and ANA-specific antibody-positive cohort ($n = 11$), anti-SSA was again the most prevalent autoantibody and was found in 5 (45.5%), followed by anti-Sm/RNP (3;27%), and anti-Ku (2;20%). While anti-Sm, anti-Ro 52, anti-Mi-2, anti-Jo-1, and anti-histones were found once.

Cytoplasmic staining was detected in eight patients of which six (75%) had ANA-specific antibodies ($P = 0.045$). Five of these were also positive for ANA. One sample with cytoplasmic staining and negative for ANA had anti-SSA antibodies.

Overall, anti-ds-DNA was done in all patients with a positive ANA where requested and was detected in 101/436 (23%) patients and in 81/190 (43%) patients with positive ANA-specific antibodies ($P < 0.001$).

Distribution of ANA patterns, anti-ds-DNA, and ANA-specific antibodies in various AID/CTD

We then analyzed the presence of ANA-specific antibodies and anti-ds-DNA with a CTD/AID diagnosis. The distribution of ANA patterns and ANA-specific antibodies with SAID is shown in Table 2. The most common SAID identified was SLE, with almost all of the patterns found in the sera of these patients except centromeric and Numa-like. The most prevalent patterns found in the sera of SLE patients were homogenous and speckled. Similarly, almost all of the ANA-specific antibodies were detected in SLE patients' sera (Table 2). Anti-ds-DNA was found in 81 SLE (62%) patients. Lupus nephritis was diagnosed in 21 patients with anti-ds-DNA in 11 (73%), anti-SSA in 12 (57%), anti-Ro 52 in 3 (14%), anti-Ku in 5 (24%), anti-histones in 9 (43%), anti-La in 6 (29%), anti-Sm in 4 (19%), anti-Sm/RNP

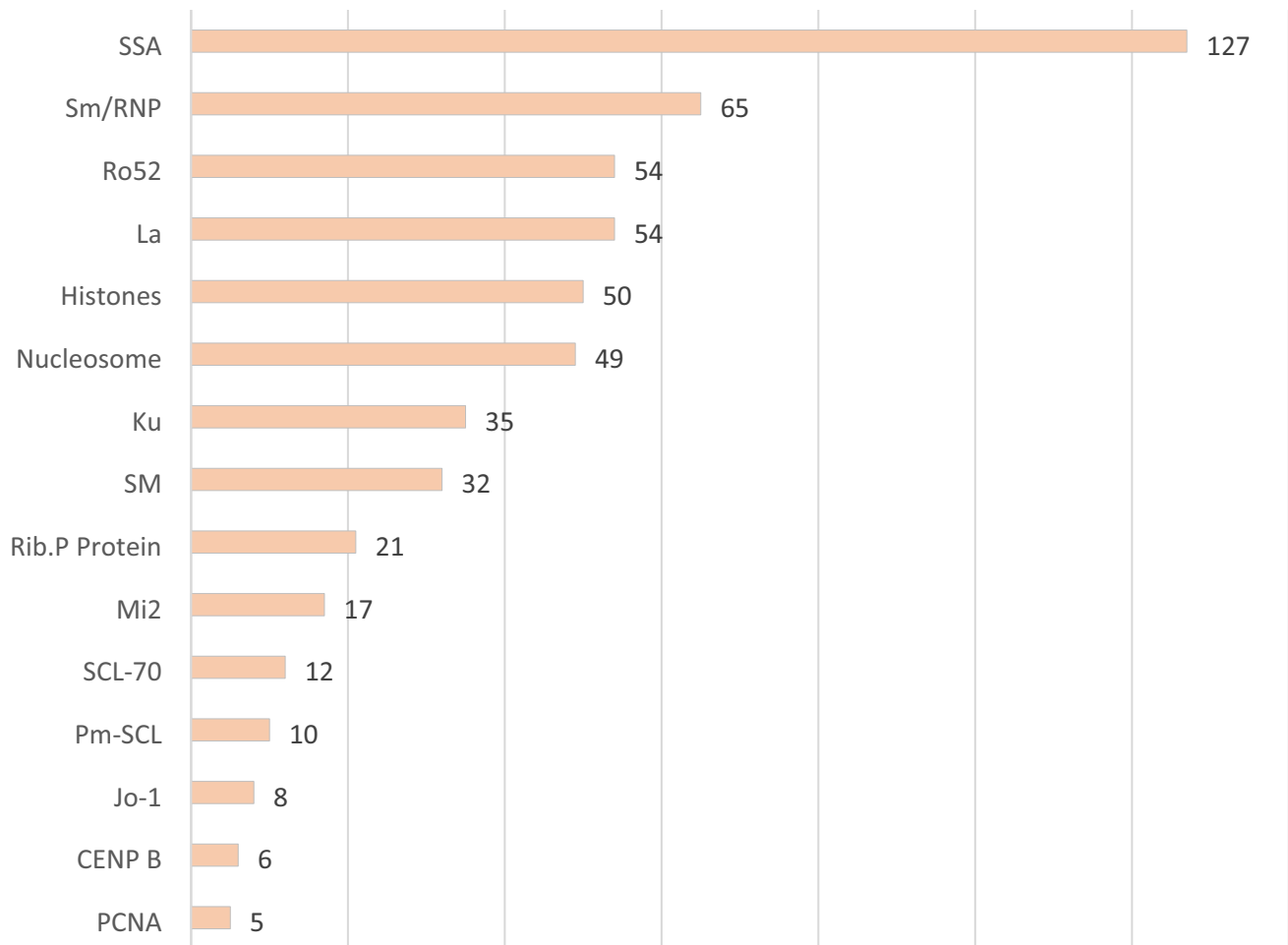


Fig. 2 Frequency distribution of ANA-specific antibodies ($n=245$)

in 7 (33%), anti-PCNA in 2 (9.5%), anti-Mi-2 in 3 (14%), and anti-Rib-p-Prt in 4 (19%).

Discussion

We correlated ANA-specific antibodies with ANA patterns in our cohort in order to specify follow-up autoantibody test cost-effectively. We found speckled or homogenous ANA patterns were the most common in our ANA-specific antibody-positive cohort as previously reported [8]. But interestingly, few ANA-specific antibodies were also seen with patterns like mitotic spindle, mid-body, or Numa-like. The autoantibodies detected in these sera were mostly anti-SSA, anti-Ro-52, and anti-SSB. As these ANA-specific antibodies do not correlate with these patterns, it may be that they correlated with negative ANA or cytoplasmic reactivity in these samples.

In the recent past, autoantibody testing for systemic rheumatic diseases has been revolutionized [10]. The integrated

approach of ANA patterns and extended panels of ANA-specific antibodies have both reduced the time-lapse in diagnosing the disease and misdiagnosis [11]. Moreover, the interpretation and reporting of ANA have been standardized to reduce subjectivity [12]. However, testing, reporting, and interpretation of these autoantibodies still pose challenges due to various factors. The testing methodologies and platforms, the expertise of immunologists and/or laboratory scientists, fluorescent microscopes variability, and pre-test probability of SAID are a few of the variables determining standards of autoantibody reporting [13, 14]. A very important factor is clinicians' understanding and acceptance of the revised reporting format of these autoantibodies. As they are familiar and more comfortable with the ANA test results and interpretation, introducing the concept of ACA with the importance of anti-cytoplasmic antibodies and anti-mitotic antibodies will require their continuous education [11, 14].

There are certain ANA patterns reported to be specifically associated with ANA-specific autoantibodies. Such as for the nucleolar pattern, ANA-specific antibodies usually

Table 1 Relationship between ANA patterns and ANA-specific antibodies

ANA-specific- antibodies (<i>n</i> = 195 ANA positives)	ANA patterns								
	Speckled AC-4, AC-5	Homogenous AC-1	Nucleolar AC-8, AC-9, AC-10	Laminar AC-11	Mitotic spindle AC-25	Dense fine speck AC-2	Centromeric AC-3	Mid-body AC-27	Numa-like AC-26
SSA (<i>n</i> = 97)	0.173	0.049	0.248**	0.032	0.045	0.010	0.103	0.010	0.073
Sm/RNP (<i>n</i> = 51)	0.225**	0.230**	0.080	0.207*	0.030	0.066	0.066	0.268**	0.121
Ro52 (<i>n</i> = 43)	0.090	0.053	0.188*	0.139	0.070	0.069	0.057	0.069	0.040
SSB (La) (<i>n</i> = 48)	0.172	0.115	0.063	0.235**	0.074	0.060	0.060	0.496**	0.042
Histones (<i>n</i> = 45)	0.430**	0.515**	0.122	0.095	0.074	0.060	0.060	0.061	0.043
Nucleosome (<i>n</i> = 43)	0.457**	0.555**	0.117	0.088	0.071	0.058	0.058	0.058	0.041
Ku (<i>n</i> = 30)	0.060	0.082	0.101	0.050	0.203*	0.050	0.050	0.050	0.035
SM (<i>n</i> = 26)	0.077	0.056	0.034	0.052	0.055	0.044	0.044	0.045	0.031
Ribosomal P protein (<i>n</i> = 17)	0.144	0.201*	0.074	0.088	0.110	0.037	0.037	0.037	0.026
Mi-2 (<i>n</i> = 14)	0.091	0.100	0.051	0.079	0.277**	0.033	0.033	0.033	0.023
Scl-70 (<i>n</i> = 10)	0.149	0.148	0.051	0.298**	0.031	0.025	0.025	0.025	0.018
Pm-Scl (<i>n</i> = 7)	0.109	0.137	0.448**	0.054	0.028	0.023	0.023	0.023	0.016
Jo-1 (<i>n</i> = 6)	0.085	0.092	0.113	0.048	0.024	0.020	0.020	0.020	0.014
CENP B (<i>n</i> = 5)	0.184*	0.070	0.293**	0.121	0.276**	0.017	0.627**	0.015	0.010
PCNA (<i>n</i> = 5)	0.053	0.062	0.034	0.040	0.020	0.017	0.017	0.017	0.012

Values considered statistically significant at **p* < 0.05 or ***p* < 0.01

Table 2 Association of ANA patterns and ANA-specific antibodies with the diagnosis of systemic autoimmune disease

	ANA-specific antibody-positives									
	SLE		MCTD		RA		Sjogren's		SSc	
	+ve	–ve	+ve	–ve	+ve	–ve	+ve	–ve	+ve	–ve
										PM/DM
ANA patterns (<i>n</i> = 195)										
Multiple patterns	8/105	10/90	6/28*	12/167	3/32	15/163	1/17	17/178	2/10	16/185
Speckled AC-4, AC-5	54/105	55/90	20/28	89/167	25/32**	84/163	15/17**	94/178	3/10	106/185
Homogenous AC-1	49/105	32/90	9/28	72/167	10/32	71/163	3/17	78/178*	6/10	75/185
Nucleolar AC-8, AC-9, AC-10	5/105	3/90	2/28	6/167	0/32	8/163	0/17	8/178	2/10**	6/185
Laminar AC-11	4/105	7/90	2/28	9/167	2/32	9/163	0/17	11/178	0/10	11/185
ANA-specific antibodies (<i>n</i> = 245)										
SSA	62/129	65/114	21/36	106/207	18/34	109/209	19/22**	108/221	6/17	121/227
Sm/RNP	44/122**	21/112	17/34**	48/200	14/34	51/200	3/22	62/212	4/16	61/218
Ro52	27/128	27/114	13/35*	41/207	7/34	47/208	6/21	48/221	5/17	49/225
La	28/129	26/113	9/36	45/206	8/34	46/208	13/21**	41/221	1/16	53/226
Histones	35/125**	15/111	1/34	49/202**	3/33	47/203	2/22	48/214	0/14	50/222*
Nucleosome	40/126**	9/113	3/35	46/204	5/34	44/205	1/22	48/217	0/15	49/224*
Ku	17/126	18/107	5/35	30/198	9/32*	26/201	3/22	32/211	1/15	34/218
SM	31/124**	1/114	4/36	28/202	8/33	24/205	2/21	30/217	0/17	32/221
Ribosomal P protein	17/127**	4/109	2/35	19/201	0/34	21/202*	2/21	19/215	1/14	20/222
Mi2	10/128	7/111	2/36	15/203	2/34	15/207	1/22	16/217	1/15	16/224
Scl-70	6/128	6/112	0/35	12/205	1/34	11/206	1/22	11/218	5/14**	7/226
Pm-Scl	4/129	6/111	3/34	7/206	3/34	7/206	1/21	9/219	4/14**	6/226
Jo-1	3/130	5/112	2/36	6/206	1/33	7/209	0/22	8/220	1/15	7/227
CENP B	3/130	3/114	0/36	6/208	0/34	6/210	0/22	6/222	0/16	6/228
PCNA	2/130	3/113	1/36	4/207	0/34	5/209	0/22	5/221	0/15	5/228

+ve, positive; –ve, negative. Chi-square test applied. Values considered statistically significant at **p* < 0.05 or ***p* < 0.01

detected are fibrillarin, PM/Scl, RNA polymerase I, etc. [6, 12, 13, 15]. On the other hand, González et al. [7] have reported a range of nuclear patterns in sera with anti-SSA, anti-SSB, and Ro 52 positivity. Rodríguez-Orozco et al. [16] have reported around 3.4% of ANA-specific antibodies positivity in patients' samples negative for ANA. With lower titers used for dilution, they were able to detect cytoplasmic staining in these samples. We have also shown the presence of certain ANA-specific autoantibodies in ANA-negative sera.

Interestingly, certain autoantibodies such as anti-ribosomal P proteins (Rib-p-prt) and anti-Jo-1 that are associated with cytoplasmic reactivity were detected in ANA-positive samples of our patients' cohort. This is also consistent with our earlier report on anti-Rib-P-Prt showing 10.5% positivity in a cohort of ANA-positive patients [17]. This shows that we cannot comfortably limit the detection of ANA-specific antibodies according to the ANA or cellular pattern observed in our patients.

According to the international consensus on ANA patterns (ICAP), it is advised to change terminology in ANA reporting to ACA on Hep-2 cells. Furthermore, the reports should mention the results of nuclear, mitotic, and cytoplasmic patterns [18]. In a Korean study, Baek et al. [19] mostly found anti-mitochondrial antibodies and anti-Rib-P-Prt in patients' sera showing cytoplasmic staining. We had reported cytoplasmic staining in a very small number of patients. As we have taken out the data from our electronic medical record system, this small number may be due to the inclusion of ANA results reported earlier before our updating of the reporting format. Future analysis on a larger sample size with cytoplasmic staining pattern with ANA-specific antibodies will enable us to look at ANA-specific antibodies correlation with greater confidence and significance. Also in this small number of samples with cytoplasmic staining, 90% of samples were positive for ANA.

According to the new algorithm of autoantibody testing for SAID as per ICAP nomenclature, ANA or ACA should be used as a screening test followed by ANA-specific antibodies or anti-ds-DNA [12]. In this study, we found that ANA was not requested for all patients for whom ANA-specific antibodies were done. One possible reason could be ANA was done in some other laboratory but the information was missing in the medical records of these patients. Nevertheless, there is a need to do a local audit and to educate our physicians regarding the proper utilization of autoantibody testing for patients with suspicion of SAID.

Another important factor is that the correct interpretation of these antibodies needs to be done in accordance with the clinical features. As we already know that most of the features of SAID are non-specific and overlapping, this makes diagnosis tricky, and initially, patients are labeled

for several SAID until proven otherwise [4, 20]. SLE is the prototype of SAID. There is a plethora of autoantibodies that can be found in SLE at varying sensitivity and specificity [21, 22]. Certain autoantibodies such as anti-SSA, anti-SSB, and anti-RNP can be found in more than one SAID [23]. In this cohort, we found positivity of almost all ANA-specific autoantibodies in SLE but limited autoantibodies' positivity in the rest of the systemic SAID. The presence of multiple patterns and multiple autoantibodies in some patients may be due to the presence of overlap syndromes. Nevertheless, investigating according to the suggested algorithm [12, 24] may help to narrow down the final diagnosis and to determine targeted therapy and appropriate prognosis accordingly [23, 25].

In our cohort, 62% of patients also had anti-ds-DNA along with ANA-specific autoantibodies. It can be argued that once a diagnosis is made due to the presence of an autoantibody then it is unnecessary to go for an extended autoantibody panel of ANA-specific autoantibodies. However, it is a known fact that certain autoantibodies are associated with a more specific symptom. Such as anti-SSA and anti-SSB are associated with congenital heart block and anti-Rib-P-Prt are associated with neuropsychiatric disorders [23]. Also, as more than one SAID can coexist [26], therefore, ANA-specific antibody testing is required even in the presence of positive anti-ds-DNA antibodies.

Conclusions

This study shows that testing for ANA-specific antibodies cannot be gated on ANA patterns. Also, there is a redundancy of these antibodies with various clinical diagnoses. Moreover, they are useful in making a diagnosis in ANA-negative patients as well with clinical suspicion.

However, following an algorithm of requesting ANA or ACA screening test for patients with suggested or justifiable suspicion of clinical features followed by anti-ds-DNA in ANA-positive patients and ANA-specific antibodies testing can be more informative and cost-effective. In case of negative ANA or unusual ANA patterns, ANA-specific antibodies should be requested only if there is a strong suspicion of SAID.

In this study, we did not find ANA patterns as a gating strategy to decide on possible ANA-specific antibody reactivity in patients' samples.

Limitations of the study

There are certain limitations of the study. ANA tests were not done for all patients. Similarly, anti-ds-DNA were also

not requested and hence was not done for all ANA-positive patients. A major limitation is clinical features of ANA-specific antibodies negative for ANA-positive patients were not analyzed. Also, the data on cytoplasmic staining and the presence of ANA-specific antibodies are limited in this study. Hence, a future study on a larger cohort of patients including results of ANA-specific antibodies in samples with all cellular patterns including nuclear, mitotic, and cytoplasmic and with clinical correlation will be more informative. There is also a need to do a more expanded analysis of clinical features in correlation with autoantibodies and rarer cellular patterns. Accordingly, using a more expanded ANA-specific antibodies panel will help us to understand the pathogenesis of the disease for targeted therapies.

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Author contribution All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by Dr. Sabiha Anis. The first and final draft of the manuscript was written by Dr. Sabiha Anis and all authors reviewed and commented on the manuscript for final version of the manuscript. All authors read and approved the final manuscript.

Declarations

Conflict of interest The authors declare no competing interests.

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